Climate Predictability and Variability in the Presence of Anthropogenic Climate Change: A Proposal for a Princeton Climate Center

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The Issues

1. The Scientific Issue: Climate Variability and Predictability.

2. The Structural Issue: A Joint Institute between GFDL and P.U.

Goals and Activities

- Experimental predictions (projections) of climate and climate change from the seasonal to the multidecadal timescales using:
  1. State-of-the-art coupled ocean-atmosphere-sea-ice models.
  2. The current state of the climate system as initial conditions.
  3. Projections of all relevant future forcing terms (GHGs, aerosols etc).
  4. Interactive terrestrial and marine carbon cycle.
- Quantification of climate predictability and variability.
- Continual development of methods for climate prediction.

Specific Products

- Periodic Reports and Assessments.
- Continually updated web site containing reports and the data from the numerical models.
- Archival publications in the scientific literature.

Limitations to Climate Predictability

1. Imperfect models;

2. Imperfect initial and boundary conditions.

Errors from these are amplified and filtered by the natural (chaotic) variability in the climate system.

Over any given timescale:

1. What processes limit predictability?

2. What level of predictability is achievable?

3. How can we achieve the maximum possible predictability?

Research Agenda

1. *Understanding and quantifying climate variability.*
   e.g., Do long-term natural oscillations (e.g., NAO, AAO) imply predictability?
   Decadal variability of ENSO.
   Understanding the historical record. Paleo-variability.

2. *Initialization.*
   Assimilation of the state of the ocean, sea-ice and (ultimately) land-surface.
   Use of Argo float program; *in situ* observations; altimeter.
   Sea-ice from satellite measurements.

3. *Radiative forcing and the Carbon Cycle.*
   Projection of levels of external forcing agents, including GHGs, sulfates and other aerosols, statistical estimates of volcanic eruptions.

   Interactive carbon cycle.
   Interaction of the terrestrial biosphere.

4. *Detection, attribution and prediction.*
   State-of-the-art models + statistical methodology.
   Differentiate between anthropogenic climate change and natural variability.

   What aspects of the climate system is predictable?
   What aspects of the initial field is the future evolution sensitive to?

Preliminary Experiments (partially completed)

- Coupled Atmosphere-Ocean GCM.
- Three different oceanic initial conditions.
- For each oceanic initial condition, three different atmospheric initial conditions.
- Nine experiments in all.

Future

- Continually perform multiple ensembles of experiments, with increasingly sophisticated models and increasingly accurate initial conditions.
- Research into the causes and nature of climate variability and predictability, including both physical and biogeochemical processes.

Structure: A Joint Institute between GFDL and PU

Phase 1: Approximately 20 additional scientists and support staff will enable the problems to be addressed.

New teams of scientists in the following areas:

1. Outreach, web-design, and interface with impacts and policy community.

2. Data Assimilation (especially ocean).

3. Statistical methodologies in prediction.

Enhancement of existing strengths in:

1. Climate variability (e.g. ‘dec-cen’ dynamics) and its quantification.

2. General model development (many aspects).

3. Radiative forcing and development of scenarios.

4. Carbon cycling, both terrestrial and marine.

Some Potential Collaborations and Interactions

1. *PMEL, AOML:* Ocean measurements, incl. Argo.

2. *NODC:* Analysed data sets. Historical record.

3. *IRI:* Impacts. Seasonal-to-interannual (S-I) to decadal.

4. *NCEP:* S-I forecasts. Development of prediction methodology.

5. *Princeton Environmental Institute,*
   Woodrow Wilson School (P.U.): Policy.

Why?

- Few (if any) major efforts in the USA to these ends. NCAR/DoE?
- There are efforts in Europe:
  e.g., Hadley Centre (U.K.). Max Planck (Germany).
- NOAA's climate services.

Why GFDL and Princeton University?

- Combined intellectual and computational resources of GFDL and Princeton University.
  - Expertise in atmospheric and oceanic modelling, and links to other NOAA labs and resources.
  - New model developments at GFDL.
  - Strength in biogeochemical cycling, paleoclimate.
  - Ability to attract and train talent.
- Commitment of existing personnel and resources.
- Grand theme with breadth, excitement and focus.